

Research Results Summary — Updated Addendum

This addendum merges the new items you requested into the existing 'Research Results Summary.pdf'. It preserves formatting, adds the updated CMD table, and appends analyses and links for the newly added and peripheral papers.

■ Updated CMD Research Results Table

Query	Total Papers	2022–Current	Mature SoA	Rigid-body	Payload	Both
Q1 – Classical / Observers	12	11	10	4	5	1
Q2 – GP / Hybrid GP	3	3	2	2	0	0
Q3 – Deep Sequence Models	9	7	6	3	2	1
Q4 – Hybrid / Residual	6	5	4	4	0	0
Q5 – Physics-Informed / Diff.	2	2	2	1	1	0
Q6 – Domain Adaptation	10	7	3	0	3	0
Total	42	35	27	14	11	2

■ Updated Mature SoA Papers (2021–2025) — Additions

Markers: **[Bib]** = in your Literature.bib, **[NEW]** = newly added.

De León et al. 2022 **[NEW]** **[Bib]**

Carreón-Díaz De León, C. L., et al. “Parameter Identification of a Robot Arm Manipulator Based on a Convolutional Neural Network.” IEEE Access, 2022.

Focus: Both rigid-body and payload. CNN maps torque/state signals to inertial parameters; more robust than LS under noise.

Link: <https://doi.org/10.1109/ACCESS.2022.3177209>

Wu et al. 2025 **[NEW]**

Wu, S., Sun, F., Chen, W., Li, Y. “Extended Deep Lagrangian Network for Robotic Arm Dynamics considering Motor Couplings.” YAC 2025.

Focus: Rigid-body estimation. Extends DeLaN to include motor couplings and nonlinear friction; improves dynamics prediction on UR10e.

Link: <https://doi.org/10.1109/YAC66630.2025.11150193>

Peng et al. 2021 **[NEW]** **[Bib]**

Peng, G., Chen, C. L. P., He, W., Yang, C. “Neural-Learning-Based Force Sensorless Admittance Control for Robots With Input Deadzone.” IEEE T-IE, 2021.

Focus: Rigid-body estimation. Sensorless external torque observer + NN controller; first NN-based admittance control with deadzone compensation.

Link: <https://doi.org/10.1109/TIE.2020.2991929>

Taie et al. 2024 ICCCR **[NEW]** **[Bib]**

Taie, W., ElGeneidy, K., Al-Yacoub, A., Sun, R. “Payload Parameters Identification Using Incremental Ensemble Learning.” ICCCR 2024.

Focus: Payload estimation. Removes need for excitation trajectories; precursor to their RA-L and Frontiers works.

Link: <https://doi.org/10.1109/ICCCR61138.2024.10585532>

■ Peripheral Papers (Not Central)

Azulay et al. 2024 **[NEW]**

Azulay, O., Mizrahi, A., Curtis, N., Sintov, A. “Augmenting Tactile Simulators with Real-like and Zero-Shot Capabilities.” ICRA 2024.

Focus: Tactile sim-to-real (GAN-based); not directly payload/rigid-body estimation.

Link: <https://doi.org/10.1109/ICRA57147.2024.10610442>

Xin et al. 2024 [NEW]

Xin, J., et al. "Programmatic Imitation Learning From Unlabeled and Noisy Demonstrations." IEEE RA-L, 2024.

Focus: Program synthesis for imitation learning; not F/T or payload estimation.

Link: <https://doi.org/10.1109/LRA.2024.3385691>